

MARK LIGHTS

Woodstock, GA • 864-980-0241 • marklights899@gmail.com
markjarredlights.com • github.com/DevMarkLights • marks-pi.com

SUMMARY

Software Engineer with 4 years of enterprise experience at General Motors, currently completing an M.S. in Computer Science (Machine Learning) at Georgia Institute of Technology. Builds and deploys production AI systems including multi-agent LLM pipelines, RAG applications, and fine-tuned language models. Strong foundation in full-stack engineering (Java, Python, React, Spring Boot) combined with hands-on AI/ML development. All AI projects are live and self-hosted.

TECHNICAL SKILLS

AI / ML	LangGraph, LangChain, LlamaIndex, ChromaDB, HuggingFace, PyTorch, Scikit-Learn, RAG, LoRA / Fine-tuning, Prompt Engineering
Languages	Python, Java, JavaScript, TypeScript, HTML, CSS, SQL
Frameworks	Spring Boot, FastAPI, React.js, Next.js, Node.js, Express.js
Cloud / Infra	Azure, Google Cloud, Docker, Cloudflare Tunneling, Raspberry Pi (self-hosted), Git
Data	Pandas, MongoDB, PostgreSQL, Spring Batch

EXPERIENCE

Software Engineer | General Motors | Roswell, GA 2022 – Present

- Led production deployments across 4 enterprise applications, coordinating with QA, DevOps, and business stakeholders to deliver releases with minimal downtime in high-availability environments
- Developed and maintained REST APIs in Spring Boot with OAuth2 authentication, integrating internal microservices and downstream systems supporting business-critical workflows
- Designed Spring Batch interfaces to optimize high-volume data processing, improving throughput and concurrency across batch jobs
- Built and tuned complex SQL queries, stored procedures, and triggers, improving query performance and ensuring data integrity across transaction-heavy workflows
- Managed P1/P2 production incident queue, performing root cause analysis and implementing long-term fixes to reduce repeat incidents and improve SLA compliance
- Delivered full-stack enhancements using Spring Boot and JSP, improving user-facing workflows across business-critical systems
- Documented production deployment and rollback procedures to support release coordination and reduce recovery time

AI / ML PROJECTS

AI Researcher — Multi-Agent Report Generation — marks-pi.com/ai-researcher/ Mar 2026 – Present

- Built a three-agent LangGraph pipeline (Planner → Researcher → Writer) that takes any question and returns a fully written, source-backed research report
- Agents coordinate autonomously: the Planner decomposes queries, the Researcher performs live web search, and the Writer synthesizes findings into structured reports
- Deployed on a self-hosted Raspberry Pi 5 with Cloudflare Tunneling — publicly accessible at marks-pi.com/ai-researcher

RAG Pipeline — Document Q&A System — marks-pi.com/rag Dec 2025 – Present

- Built a full-stack retrieval-augmented generation system allowing users to upload PDF and TXT files and query their contents via semantic search
- Implemented Python/FastAPI backend using all-MiniLM-L6-v2 embeddings and ChromaDB for vector storage and similarity search
- Built React/Express frontend and self-hosted the full stack on Raspberry Pi 5

Local LLaMA Voice Assistant — github.com/DevMarkLights/Ai-Assistant May 2025 – Jun 2025

- Deployed LLaMA 3.2 Instruct locally and built a voice-enabled interface with real-time speech-to-text and text-to-speech for hands-free conversational AI
- Optimized for low-latency on-device inference — fully private, no external API calls

Lights Finance — Portfolio Analytics App — marks-pi.com/finance Nov 2023 – Nov 2025

- Built a finance web app analyzing portfolio performance, sector diversification, dividend income, and options rate-of-return
- Integrated a custom Yahoo Finance REST API (Flask) for real-time stock and options data, and a Robinhood report parser for monthly income calculations
- Self-hosted on Raspberry Pi with Cloudflare Tunneling

EDUCATION

M.S. Computer Science — Machine Learning | [Georgia Institute of Technology](#) | Jan 2024 – Present

- Relevant coursework: Machine Learning (CS7641), Deep Learning (CS7643), Natural Language Processing / LLMs (CS7650), Artificial Intelligence (CS6601), ML for Trading (CS7646)
- Built a large language model from scratch using PyTorch with LSTM/RNN architecture; implemented transformers and CNNs in Deep Learning
- Trained a reinforcement learning model for stock trading strategy optimization in ML for Trading

B.S. Computer Science | [Francis Marion University](#) | Aug 2015 – May 2022

RESEARCH

Fine-Tuning Research Paper — Published research on fine-tuning large language models using parameter-efficient methods (LoRA/PEFT). <https://markjarredlights.com/articles>